

ALGORITHM

Spring 2026, Quiz 6, April 8

11:45-12:05 am

姓名: 黃翊宏

學號: 01357101

系所: CS 2B

1. What is the running time of HEAPSORT on an array A of length n that is already sorted in increasing order? What about decreasing order?

2. Illustrate the operation of MAX-HEAPIFY($A, 2$) on the array $A = \langle 28, 2, 13, 15, 12, 9, 0, 4, 6, 11, 3, 7, 8 \rangle$. (1-based)

1. increasing order = decreasing order running Time = $O(n \lg n)$

∴ 每次插入新值時，都要先把值塞進最下方的 leaf node，再根據 Heap 是 max-heap 還是 min-heap 決定 call increase-key 還是 decrease-key 上浮

∴ running time 都是 $O(n \lg n)$

2. max-heapify(A, i)

$l \leftarrow \text{left}(i)$

$r \leftarrow \text{right}(i)$

if $i > 1$ and $A[l] > A[i]$

then $\text{largest} \leftarrow l$

else $\text{largest} \leftarrow i$

if $i > 1$ and $A[r] > A[\text{largest}]$

then $\text{largest} \leftarrow r$

if $\text{largest} \neq i$

do exchange $A[i] \leftrightarrow A[\text{largest}]$

max-heapify($A, \text{largest}$)

ORIG:

